

Assignment # 01

Signals and Systems

Name:

Ahmad Sheraz

Roll no:

21L-5720

Section:

BEE-4A

Submitted To:-

Sir Shaikh Muhammad Sajid

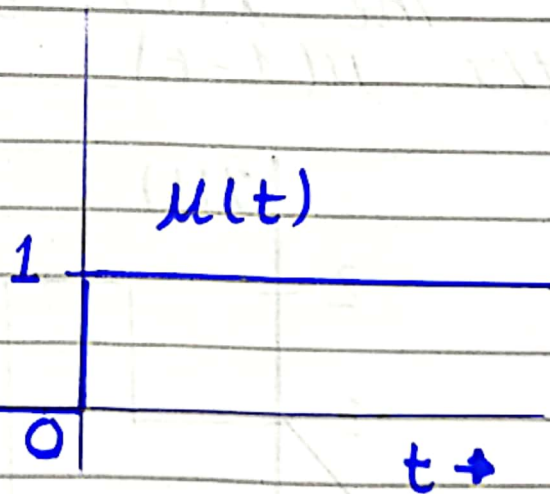
Question #01

Sketch and label each of the following

a) $x(t)u(1-t)$

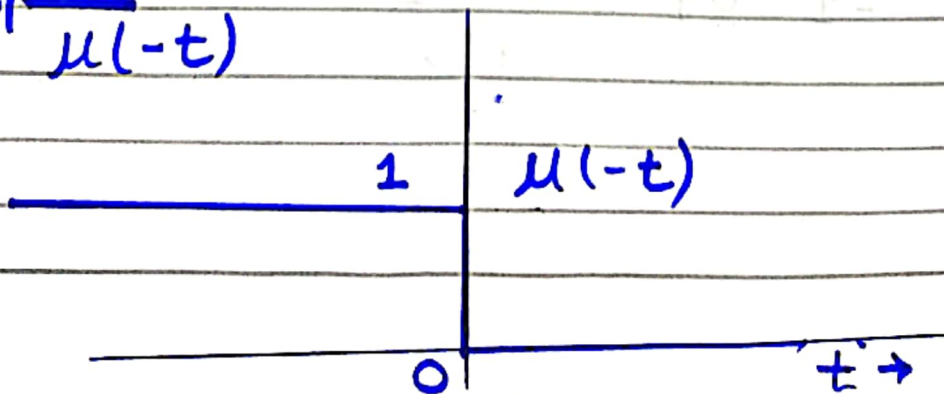
Step 1:-

$\rightarrow u(t)$



Step 2:-

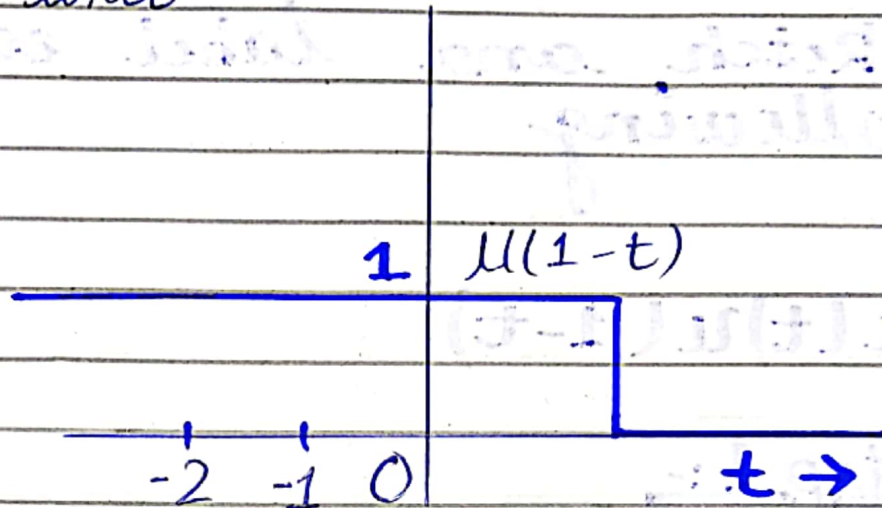
$u(1-t)$



Step 3:

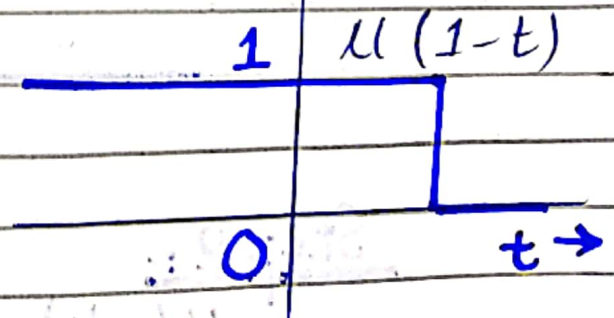
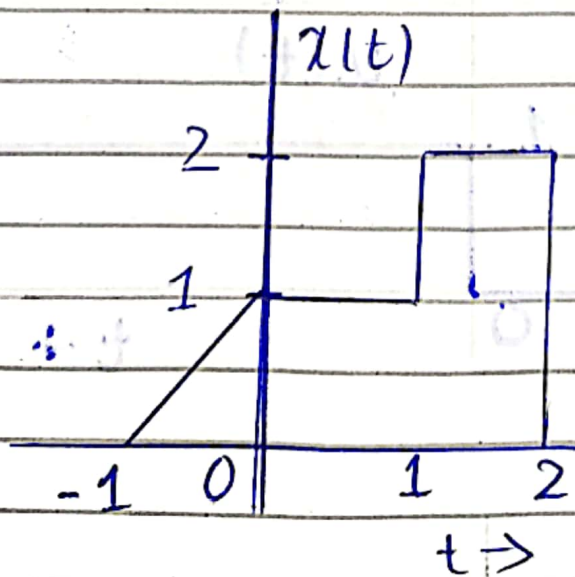
$$\mu(1-t)$$

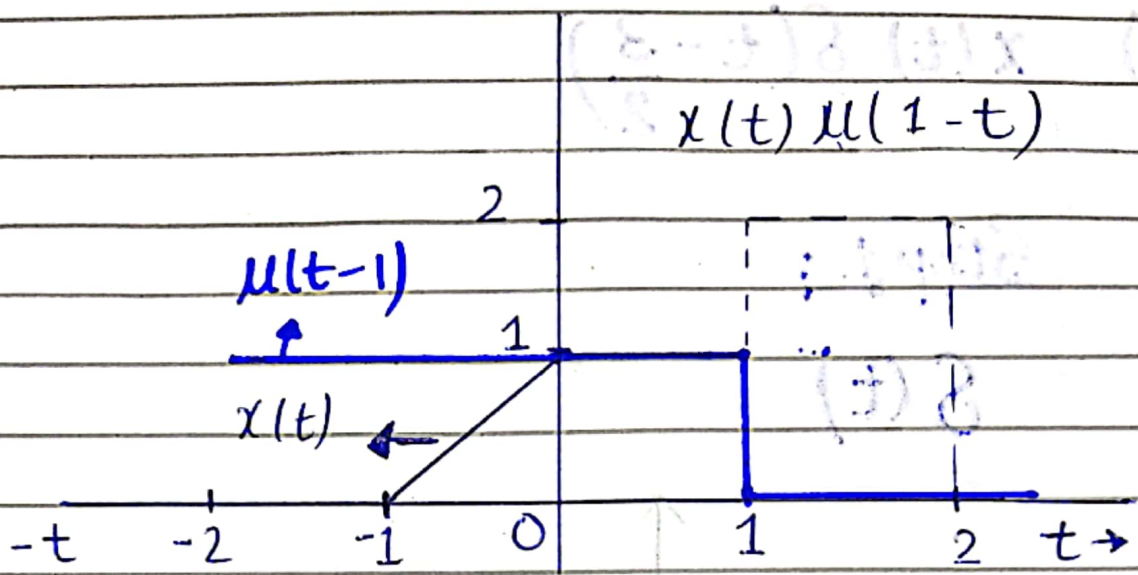
→ The graph will shift rightwards by 1 unit



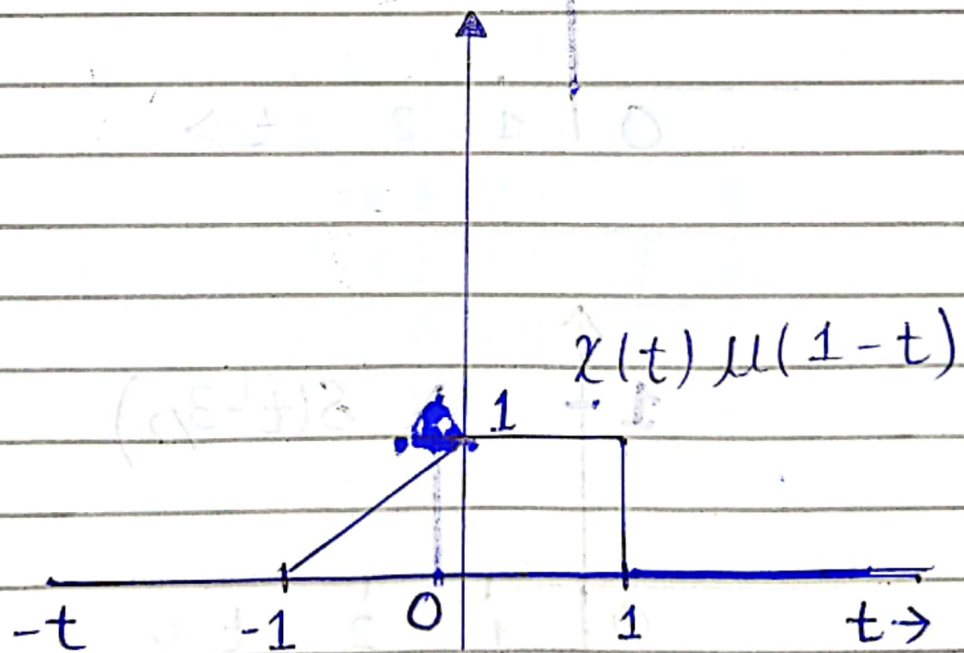
Step 4:

The $x(t)$ will pass through the $\mu(1-t)$





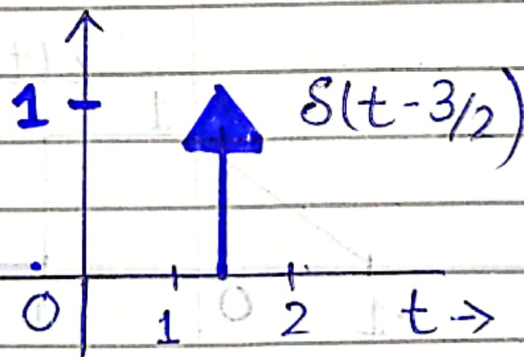
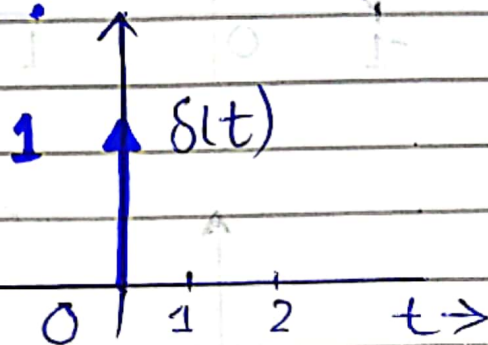
$\Rightarrow$



b) $x(t) \delta(t - \frac{3}{2})$

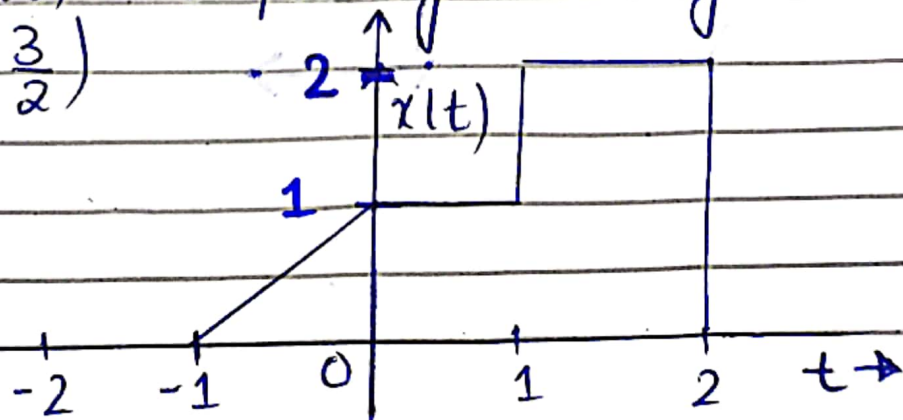
Step 1:

$\delta(t)$

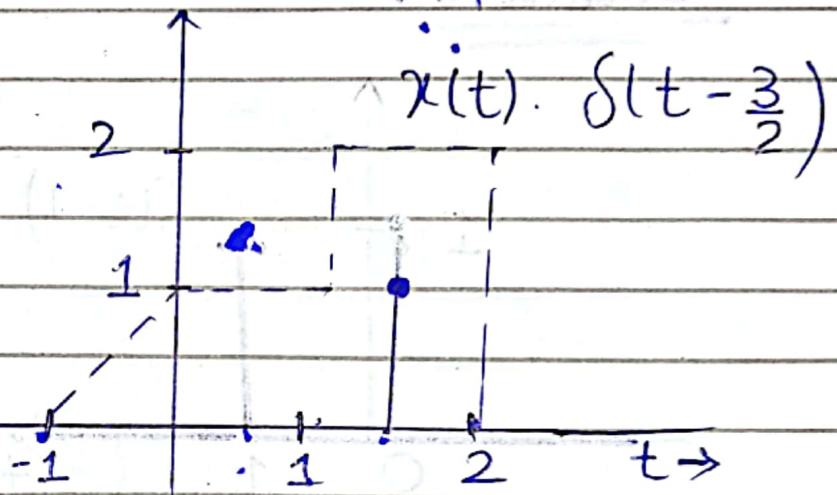
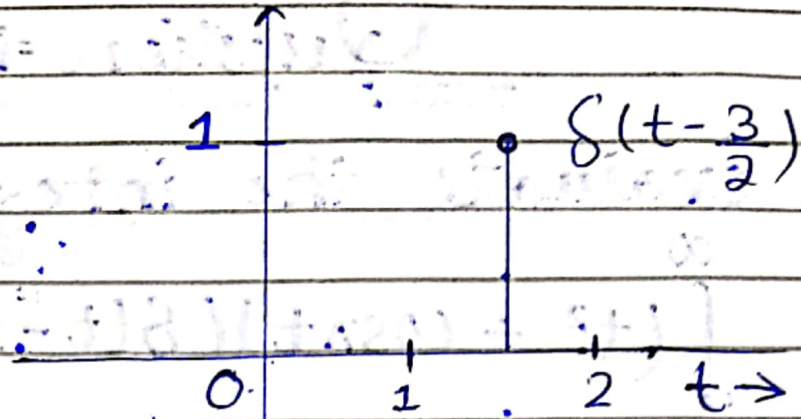


Step 2:-

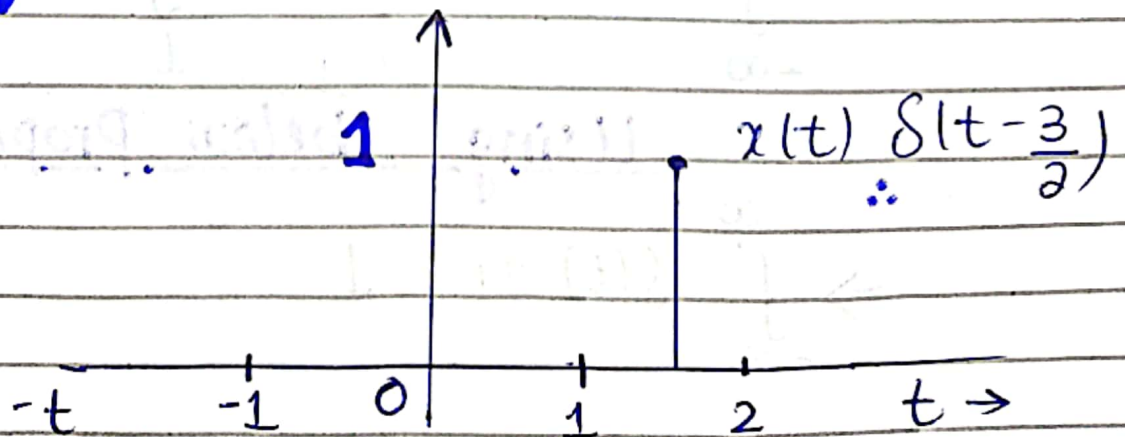
$x(t)$ is passing through the $\delta(t - \frac{3}{2})$



⇒



Therefore;

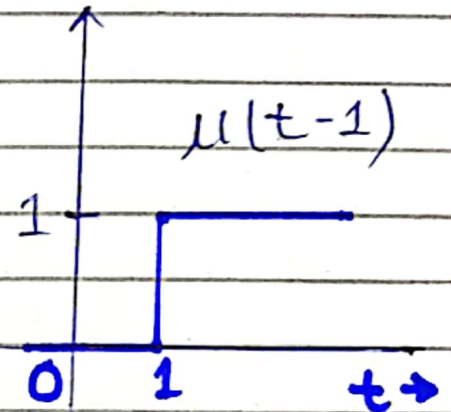
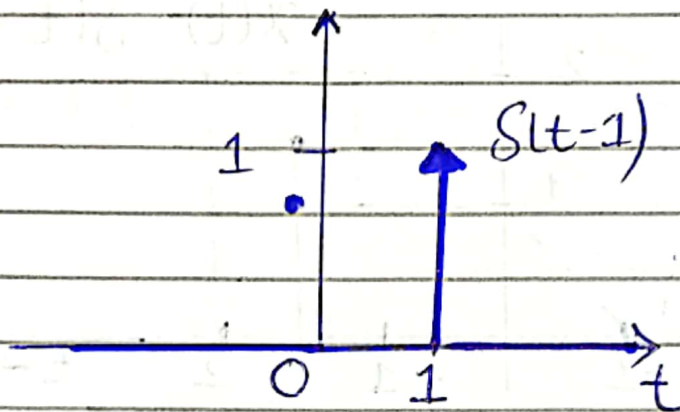


Question #02

Evaluate the integrals

a) $\int_{-\infty}^{\infty} (t^2 + \cos \pi t) (\delta(t-1)) dt$

Solution :-



$$= \int_{-\infty}^1 (t^2 + \cos \pi t) \cdot 0 dt + \int_1^{\infty} (t^2 + \cos \pi t) \cdot 1 dt =$$

Using below Properties

$$\therefore \rightarrow \int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\rightarrow \int_{-\infty}^{\infty} \phi(t) \delta(t-T) dt = \phi(T)$$

where;

$T=1$ "in above case"

$\Rightarrow \phi(t)$ is continuous time function

Therefore;

$$= \int_{-\infty}^{\infty} (t^2 + \cos \pi t) \delta(t-1) dt$$

→ "By using Property of unit Impulse function"

$$= (t^2 + \cos \pi t)_{t=T} \quad \therefore \phi(t) \Rightarrow \phi(T)$$

$$= (t^2 + \cos \pi t)_{t=1}$$

$$= ((1)^2 + \cos \pi \times 1)$$

$$= (1 - 1)$$

$$= 0$$



$$b) \int_{-\infty}^{\infty} (e^{-t}) \delta(2t-2) dt$$

Solution:-

$$\int_{-\infty}^{+\infty} e^{-t} \delta(2t-2) dt$$

$$\therefore \int_{-\infty}^{\infty} \phi(t) \delta(t-T) dt = \phi(T)$$

Therefore;

$$\int_{-\infty}^{+\infty} e^{-t} \delta(2t-2) dt = e^{-t} \Big|_{t=2}$$

$$\int_{-\infty}^{\infty} e^{-t} \delta(2t-2) dt = e^{-T}$$

→ where, $T=2$

$$\Rightarrow \int_{-\infty}^{\infty} e^{-t} \delta(2t-2) dt = e^{-2} = 0.135$$

Question # 03

a) $x(t) = u(t) - u(t-6)$

Find and sketch the first derivative of following signal

Solution :-

$$x(t) = u(t) - u(t-6)$$

Sampling Property of $\delta(t)$:-

$$\therefore \left(\frac{du}{dt} = \delta(t) \right)$$

Therefore:

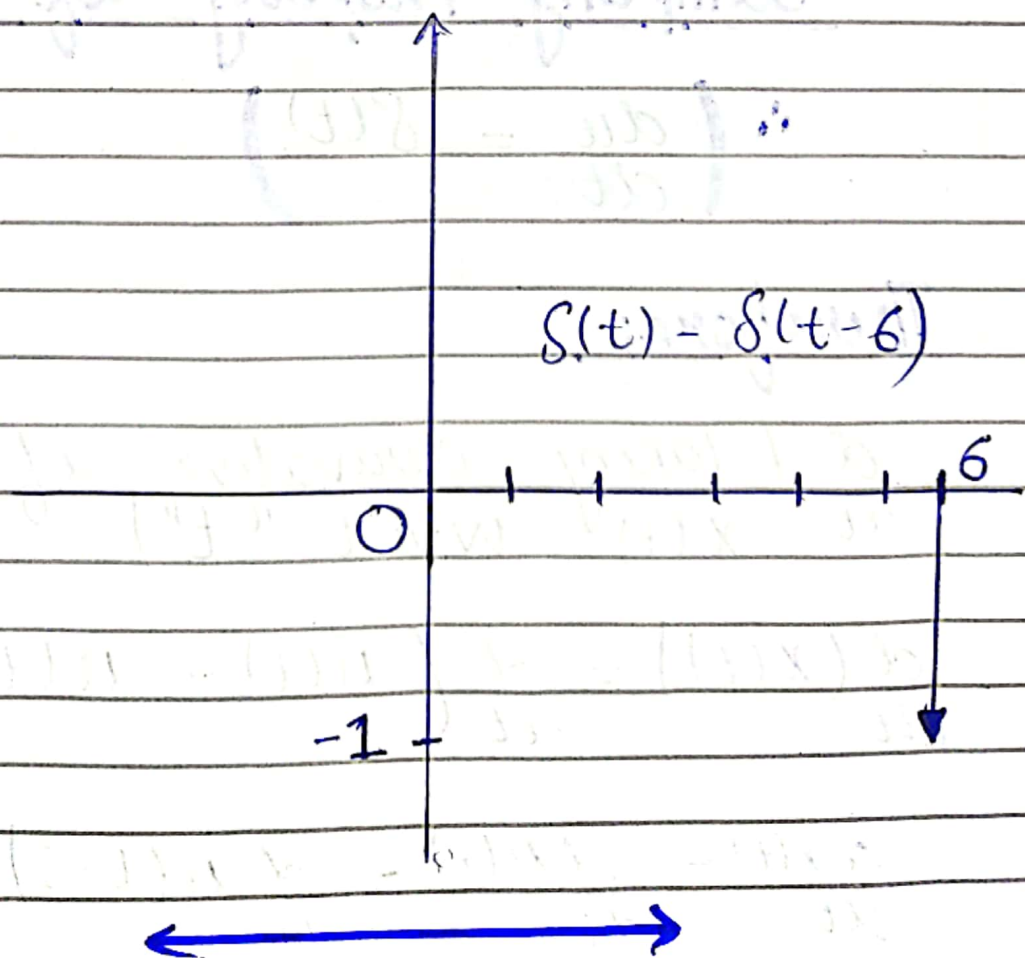
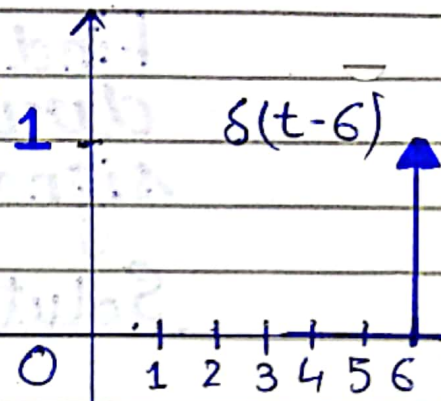
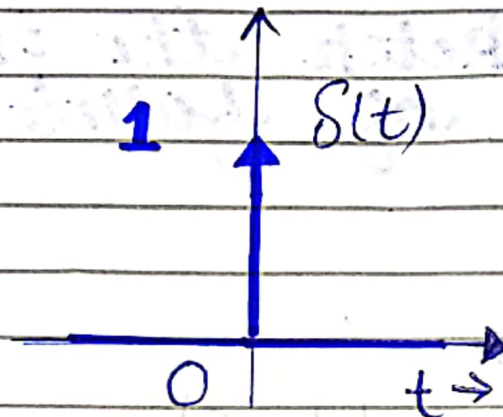
$\frac{d}{dt} (x(t))$ (Taking derivative of signal w.r.t "t")

$$\frac{d}{dt} (x(t)) = \frac{d}{dt} (u(t) - u(t-6))$$

$$\frac{dx(t)}{dt} = \frac{du(t)}{dt} - \frac{du(t-6)}{dt}$$

$$\Rightarrow \frac{dx(t)}{dt} = \delta(t) - \delta(t-6)$$

Sketching above first derivative:-



b) $x(t) = t [u(t) - u(t-6)]$

Solution:-

$\Rightarrow x(t) = t [u(t) - u(t-6)]$

$\frac{d}{dt} x(t) = \frac{d}{dt} [t (u(t) - u(t-6))]$

$\frac{d}{dt} x(t) = t \left[\frac{d}{dt} u(t) - \frac{d}{dt} u(t-6) \right]$

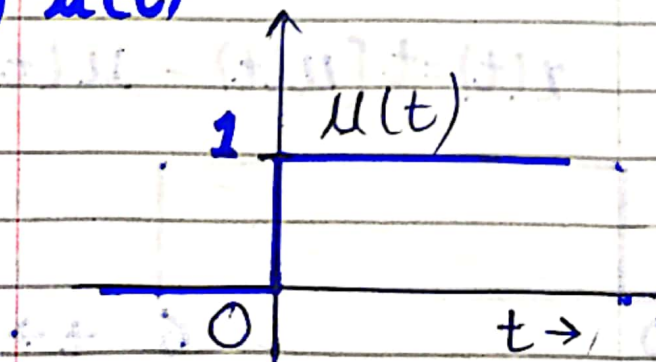
$\frac{dx(t)}{dt} = t [\delta(t) - \delta(t-6)]$

$\rightarrow$ Sketching the original signal $x(t)$:-

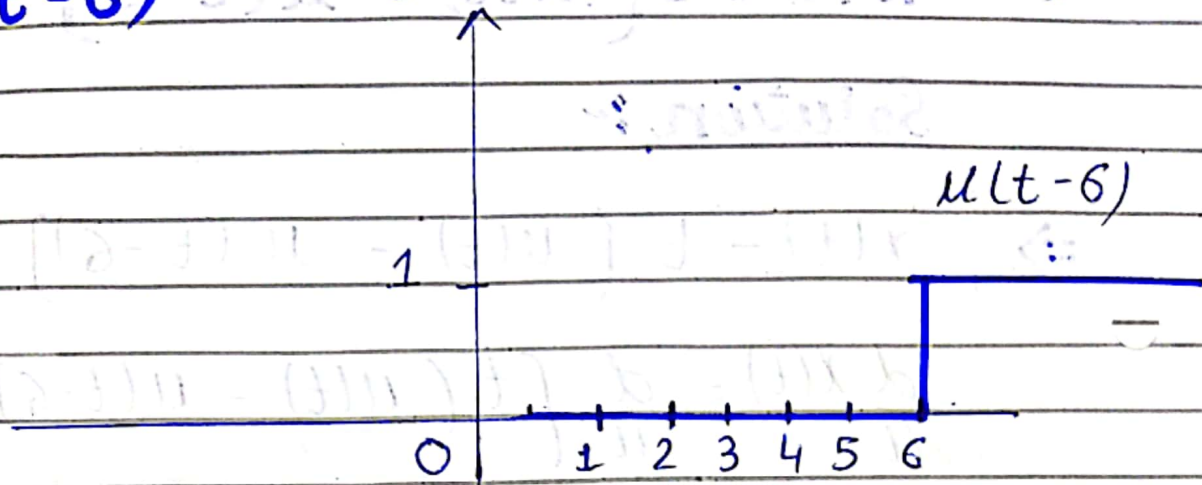
$x(t) = t u(t) - t u(t-6)$

$x(t) = t [u(t) - u(t-6)]$

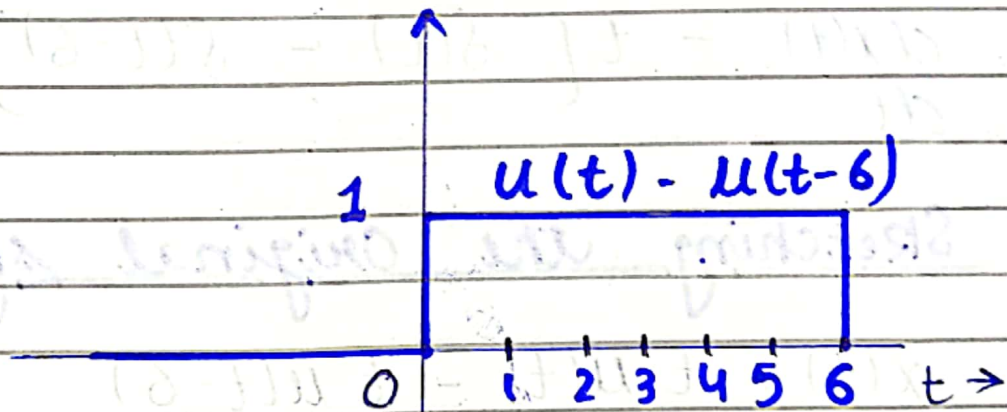
.) $u(t)$



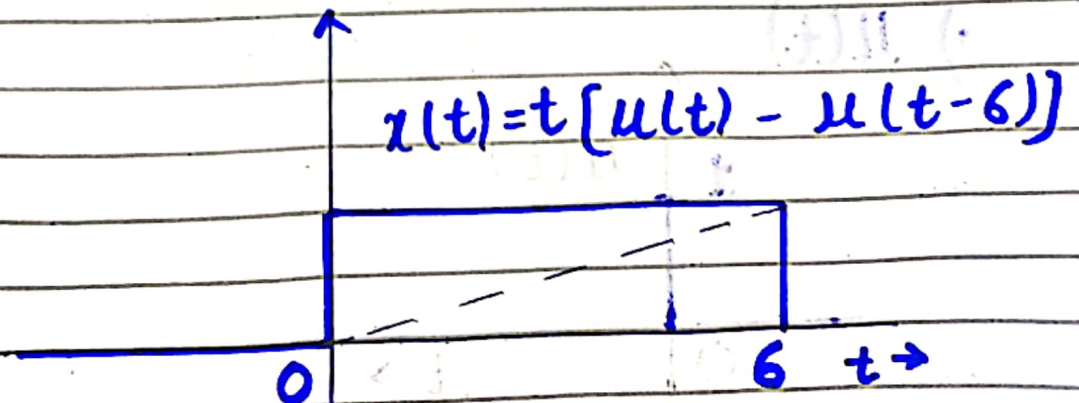
.) $u(t-6)$



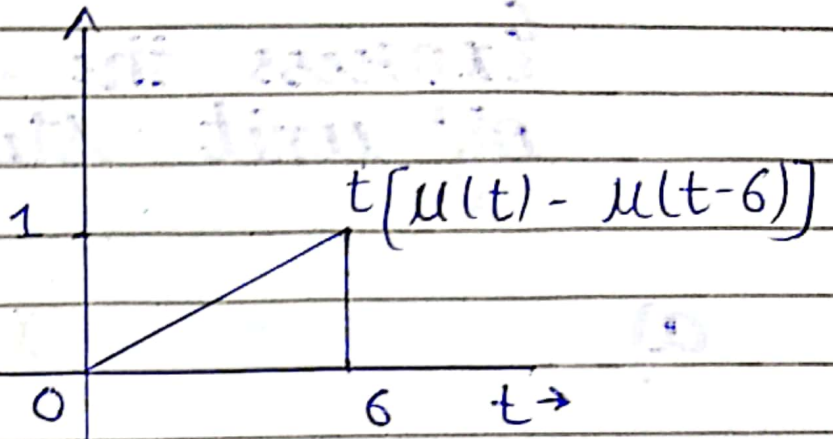
gate Pulse :-
 $u(t) - u(t-6)$



Now, $t[u(t) - u(t-6)]$

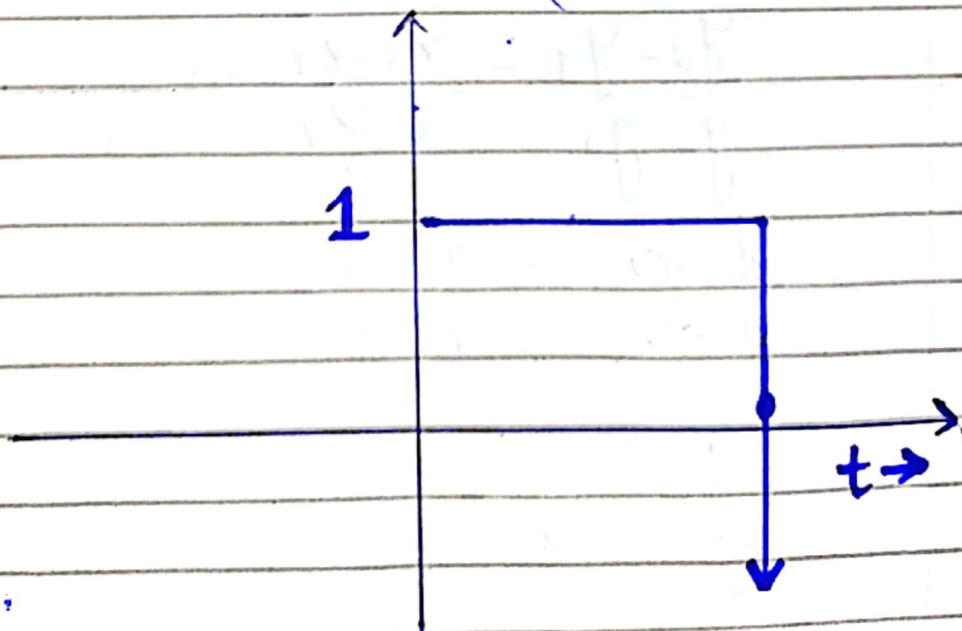


Therefore;



where derivative will :-

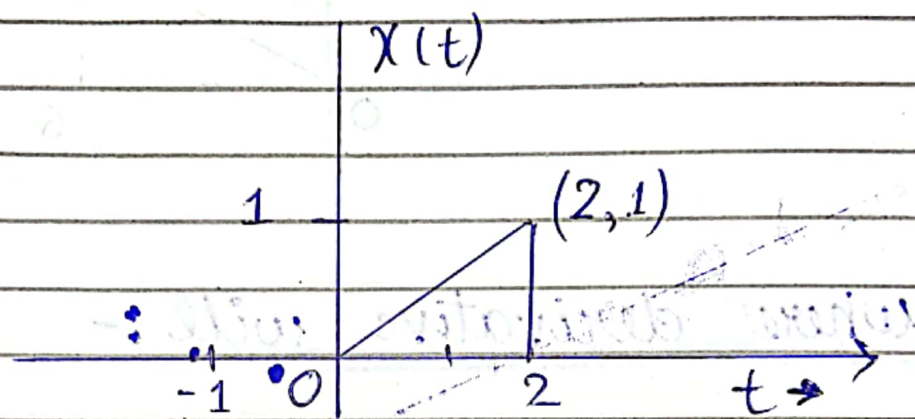
$$\frac{dx(t)}{dt} = t[\delta(t) - \delta(t-6)]$$



Question # 04

Express the signals in terms of unit step function.

a)



The equation of the ramp is;

$$\frac{y_2 - y_1}{y_2 - y_1} = \frac{x_2 - x_1}{x_2 - x_1}$$

$$\frac{1 - 0}{y - 0} = \frac{2 - 0}{x - 0}$$

$$\therefore \frac{1}{y} = \frac{2}{x}$$

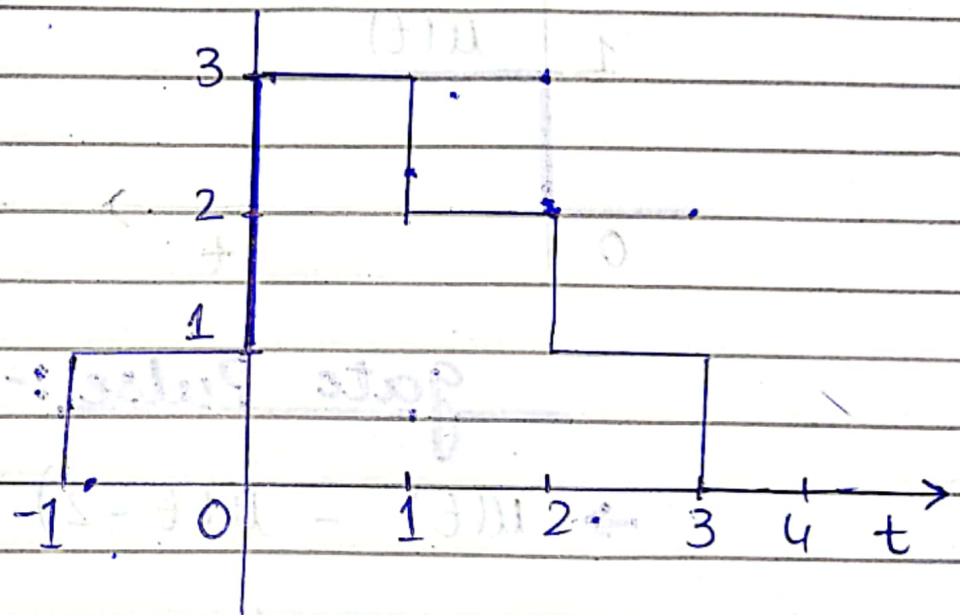
$$\frac{x}{2} = y \Rightarrow \boxed{y = \frac{t}{2}}$$

Therefore;

$$x(t) = \frac{t}{2} \left(u(t) - u(t-2) \right)$$

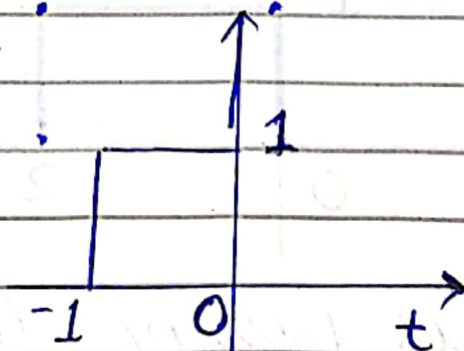


b)



Solution :-

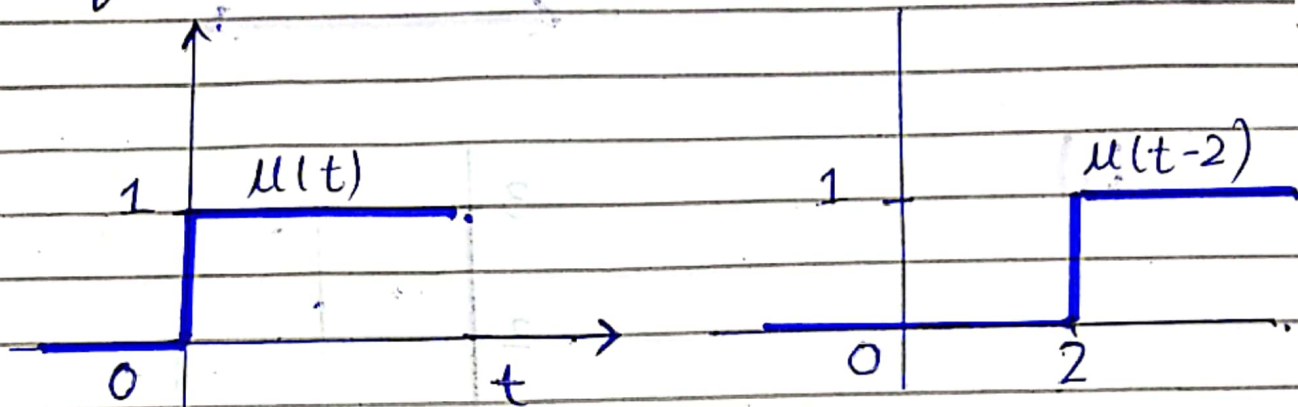
→ Step 1 :-



$$x_1(t) = 1 \left[u(t+1) - u(t) \right]$$

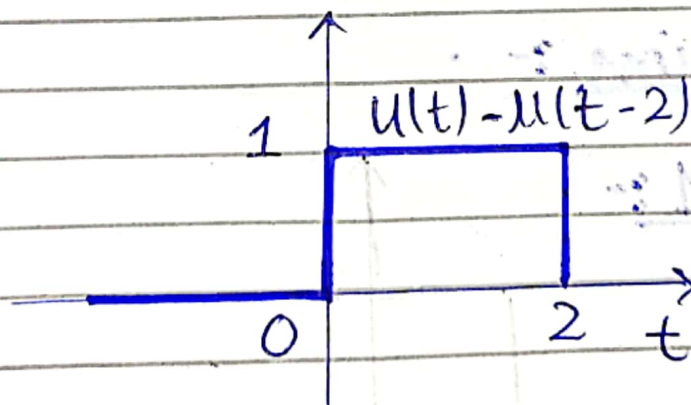
The first point is $(0,0)$ and second is $(2,1)$

Therefore;

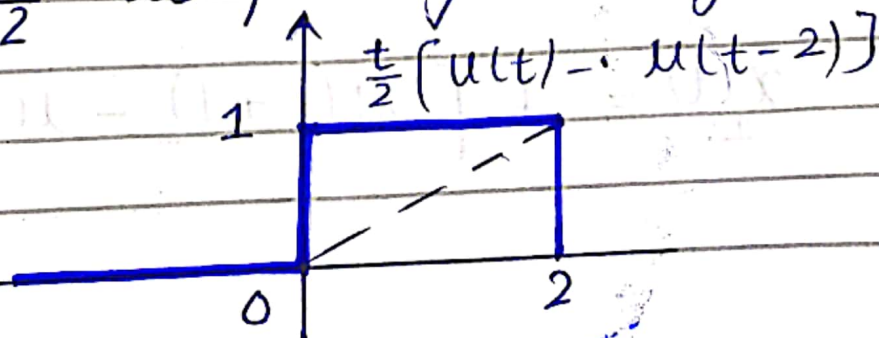


gate Pulse :-

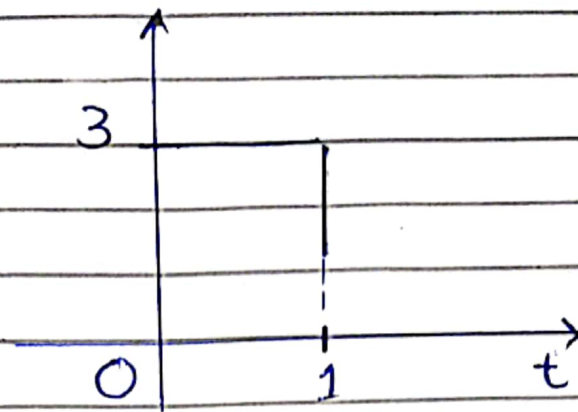
$$\rightarrow u(t) - u(t-2)$$



The $\frac{t}{2}$ is passing through it

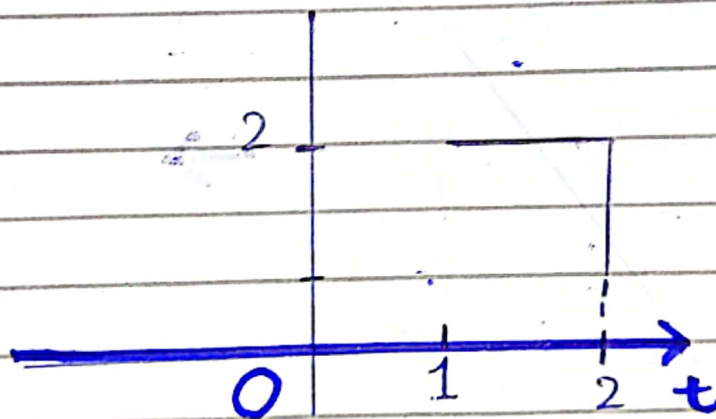


Step 2:-



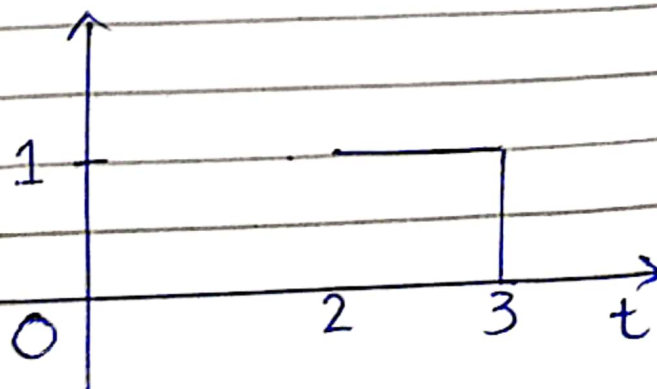
$$x_2(t) = 3 [u(t) - u(t-1)]$$

Step 3:-



$$x_3(t) = 2 [u(t-1) - u(t-2)]$$

Step 4:-



$$x_4(t) = 1[u(t-2) - u(t-3)]$$

Now;

$$x(t) = x_1(t) + x_2(t) + x_3(t) + x_4(t)$$

$$x(t) = 1[u(t+1) - u(t)] + 3[u(t) - u(t-1)] + 2[u(t-1) - u(t-2)] + 1[u(t-2) - u(t-3)]$$

$$x(t) = u(t+1) - u(t) + 3u(t) - 3u(t-1) + 2u(t-1) - 2u(t-2) + 1u(t-2) - u(t-3)$$

$$x(t) = u(t+1) - u(t)[1-3] - u(t-1)[3-2] - u(t-2)[2-1] - u(t-3)$$

$$x(t) = u(t+1) + 2u(t) - u(t-1) - u(t-2) - u(t-3)$$



Question # 05

→ Find and sketch the odd and even component of $\sin \omega_0 t u(t)$

∴ Solution:-

As we know; every signal $x(t)$ can be expressed as sum of even and odd components, because

$$x(t) = \underbrace{\frac{1}{2} [x(t) + x(-t)]}_{\text{even}} + \underbrace{\frac{1}{2} [x(t) - x(-t)]}_{\text{odd}}$$

as $x(t) = \sin \omega_0 t u(t)$

∴ Even Component:-

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$x_e(t) = \frac{1}{2} [\sin \omega_0(t) u(t) + \sin \omega_0(-t) (u(-t))]$$

1) Odd Component :-

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)]$$

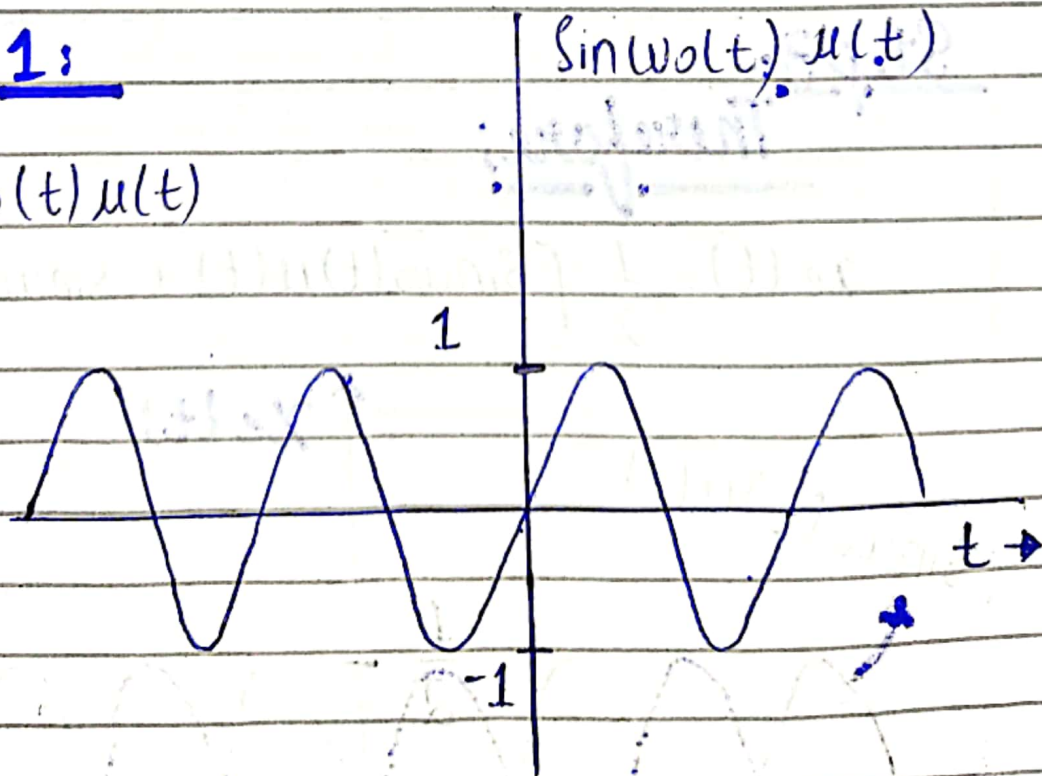
$$x_o(t) = \frac{1}{2} [\sin \omega_0 t) u(t) - \sin \omega_0 (-t) u(-t)]$$

→ Sketching even component ($x_e(t)$) :-

$$x_e(t) = \frac{1}{2} [\sin \omega_0 t) u(t) + \sin \omega_0 (-t) u(-t)]$$

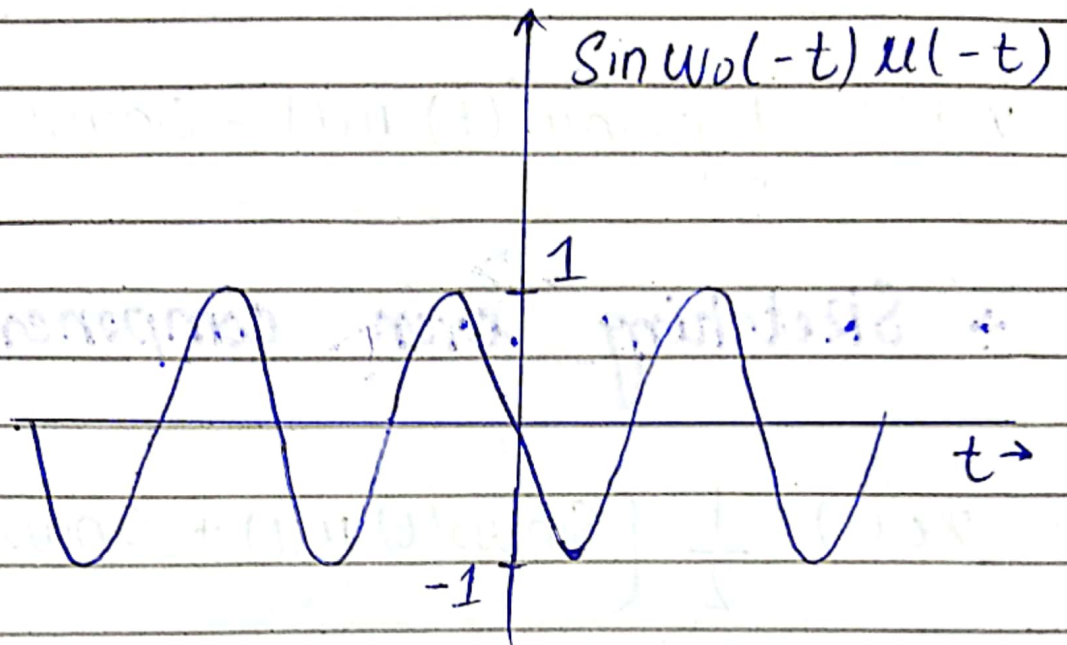
Step 1:

$\sin \omega_0 t) u(t)$



Step 2:

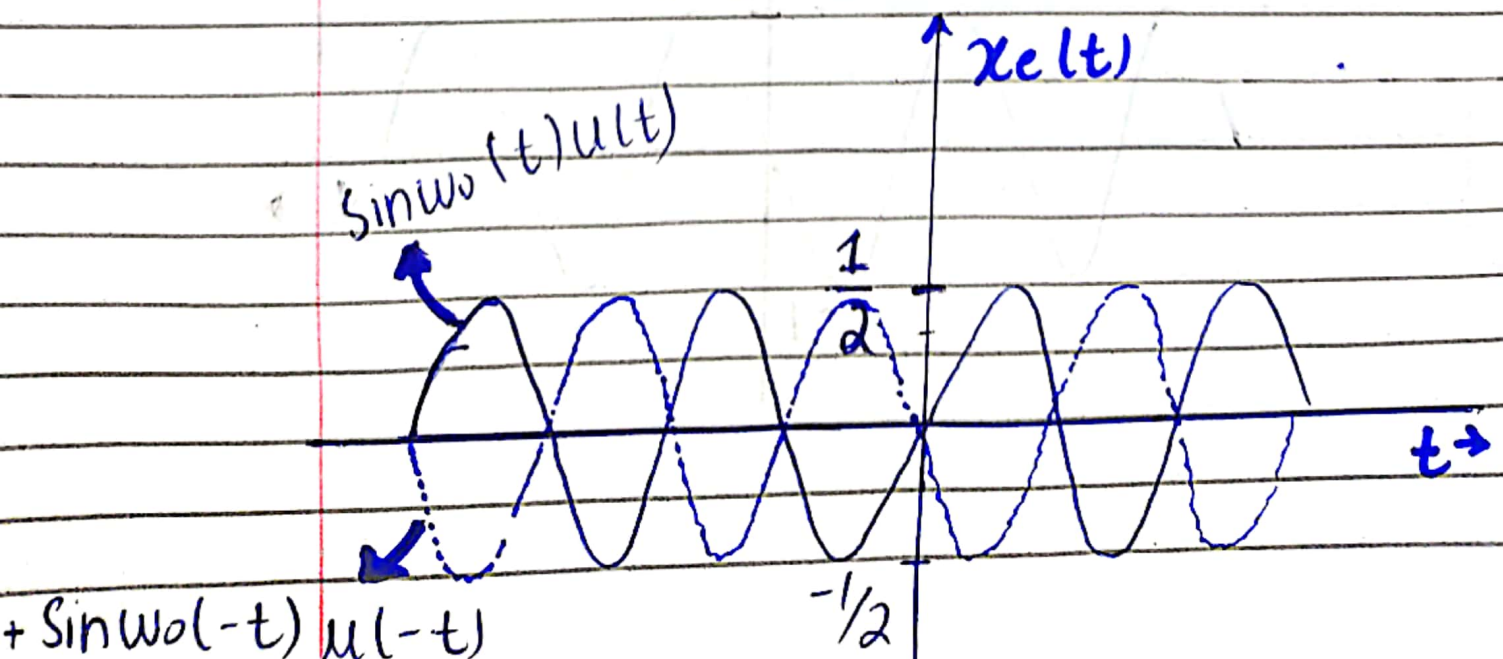
$$\sin \omega_0(-t) u(-t)$$



Step 3:-

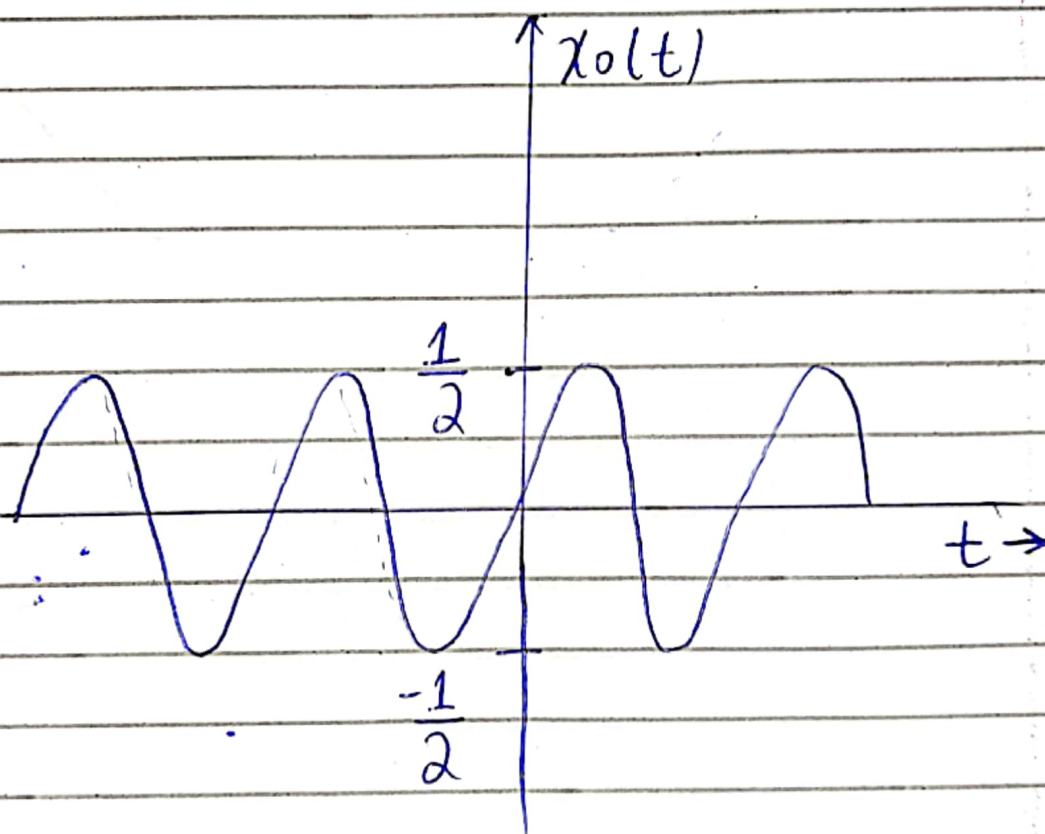
Therefore;

$$x_e(t) = \frac{1}{2} [\sin \omega_0(t) u(t) + \sin \omega_0(-t) u(-t)]$$



→ Sketching the odd component $x_o(t)$

$$x_o(t) = \frac{1}{2} \left[\sin \omega_0 t u(t) - \sin \omega_0 (-t) u(-t) \right]$$



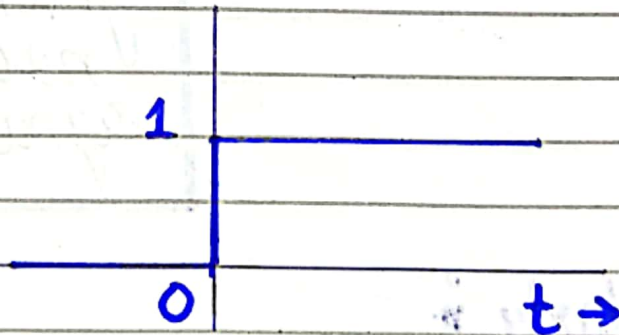
Question #06

Which of the following are energy and power signals or neither?

Solution

a) $x(t) + 5x(t-1) - 2x(t-2)$

$\Rightarrow x(t)$



$$x(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

$\rightarrow$ Energy (E_x):-

$$E_x = \int_{-\infty}^{\infty} x^2(t) dt$$

$$E_x = \int_{-\infty}^0 0^2 dt + \int_0^{\infty} 1^2 dt$$

$$E_x = 0 + \int_0^{\infty} 1 dt$$

$$E_x = [t]_0^{\infty}$$

$$E_x = \infty - 0 = \infty$$

Therefore $x(t)$ is not a energy signal. and the given signal is not a energy signal

Similarly :-

Power (P_x) :-

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (1)^2 dt$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \left[t \right]_0^T$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} [T - 0]$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} [T]$$

$$P_x = \lim_{T \rightarrow \infty} (1)$$

$$\boxed{P_x = 1}$$

so $u(t)$ is a power function
 Therefore given signal is a
power signal

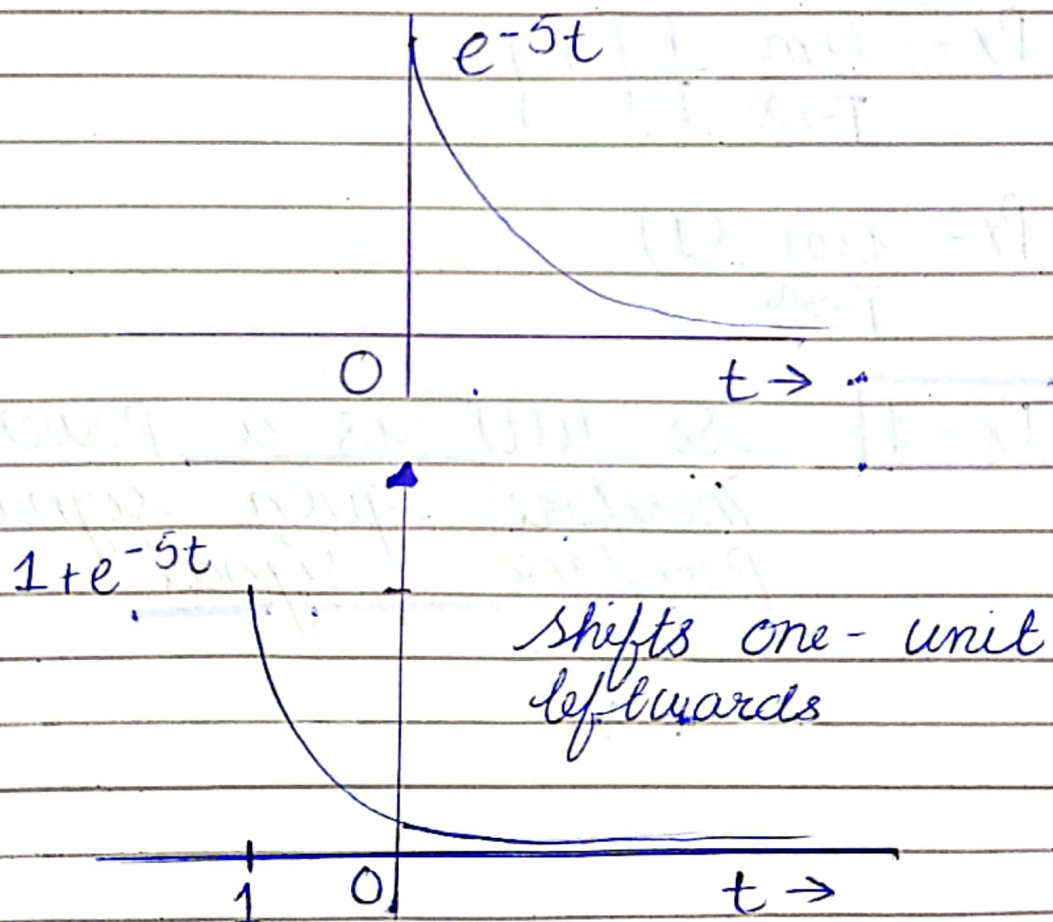


$$b) (1 + e^{-5t}) u(t)$$

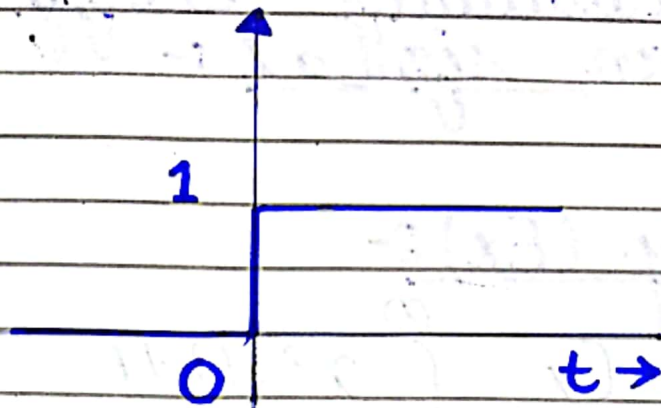
Solution:-

$$(1 + e^{-5t}) u(t)$$

$$\Rightarrow 1 + e^{-5t}$$

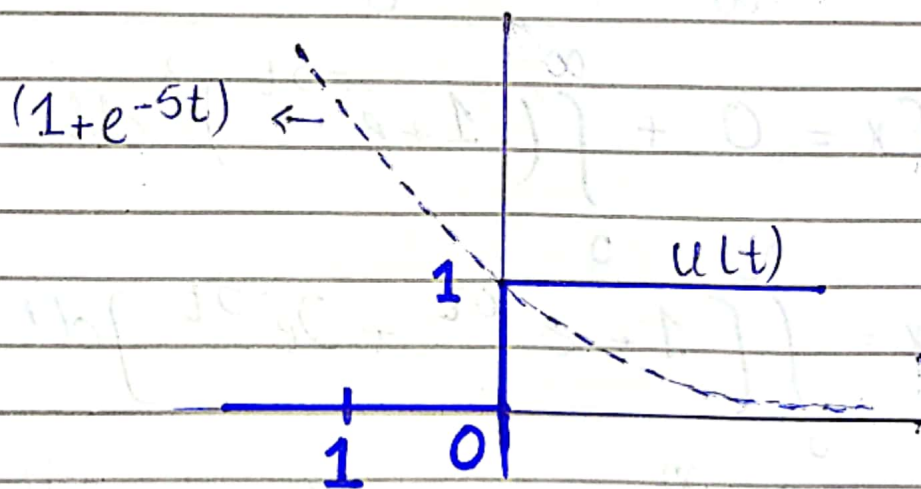


$$\Rightarrow u(t)$$

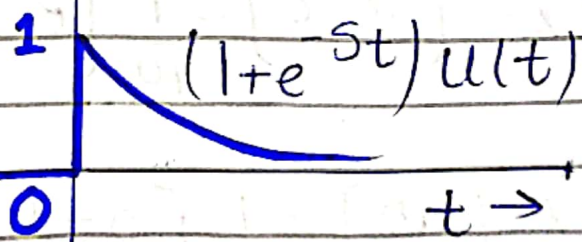


Therefore

$$(1 + e^{-5t})u(t)$$



$\Rightarrow$



Now calculating energy and Power for given signal :-

Energy (E_x) :-

$$E_x = \int_{-\infty}^{\infty} x^2(t) dt$$

$$E_x = \int_{-\infty}^0 0^2 dt + \int_0^{\infty} ((e^{-5t} + 1) u(t))^2 dt$$

$$E_x = 0 + \int_0^{\infty} (1 + e^{-5t})^2 dt$$

$$E_x = \int_0^{\infty} [1 + e^{-10t} + 2e^{-5t}] dt$$

$$E_x = t \Big|_0^{\infty} + \frac{e^{-10t}}{-10} \Big|_0^{\infty} + 2 \frac{e^{-5t}}{-5} \Big|_0^{\infty}$$

$$E_x = [\infty - 0] - \frac{1}{10} [e^{-\infty} - e^0] - \frac{2}{5} [e^{-\infty} - e^0]$$

$$E_x = [\infty] - \frac{1}{10} [0 - 1] - \frac{2}{5} [0 - 1]$$

$$E_x = \infty + \frac{1}{10} + \frac{2}{5}$$

$$E_x = \infty$$

Therefore, the given signal is not a energy signal.

Power (P_x):-

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \left[\int_0^T ((1 + e^{-5t}) u(t))^2 dt \right]$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \left[\int_0^T (1 + e^{-5t})^2 dt \right]$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \left[\int_0^T (1 + e^{-10t} + 2e^{-5t}) dt \right]$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \left[t \Big|_0^T + \frac{e^{-10t}}{-10} \Big|_0^T + 2 \frac{e^{-5t}}{-5} \Big|_0^T \right]$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \left[T - \left[\frac{e^{-10T} - 1}{10} \right] - \frac{2}{5} \left[\frac{e^{-5T} - 1}{1} \right] \right]$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \left[T - \frac{1}{10} [e^{-10T} - 1] - \frac{2}{5} [e^{-5T} - 1] \right]$$

$$P_x = \lim_{T \rightarrow \infty} \left[1 - \frac{1}{10T} [e^{-10T} - 1] - \frac{2}{5T} [e^{-5T} - 1] \right]$$

$$P_x = \lim_{T \rightarrow \infty} 1 - \lim_{T \rightarrow \infty} \frac{1}{10T} [e^{-10T} - 1]$$

$$- \lim_{T \rightarrow \infty} \frac{2}{5T} [e^{-5T} - 1]$$

$$P_x = 1 - \frac{1}{10(\infty)} [e^{-\infty} - 1] - \frac{2}{5(\infty)} [e^{-5(\infty)} - 1]$$

$$P_x = 1 - 0(0 - 1) - 0(0 - 1)$$

$$\boxed{P_x = 1}$$

Hence as $P_x = 1$ that is why the the given signal is Power signal

Conclusion:-

→ The Power signals are the ones whose total power is finite and **non-zero**. A sinusoidal signal has finite, non-zero power but infinite energy.

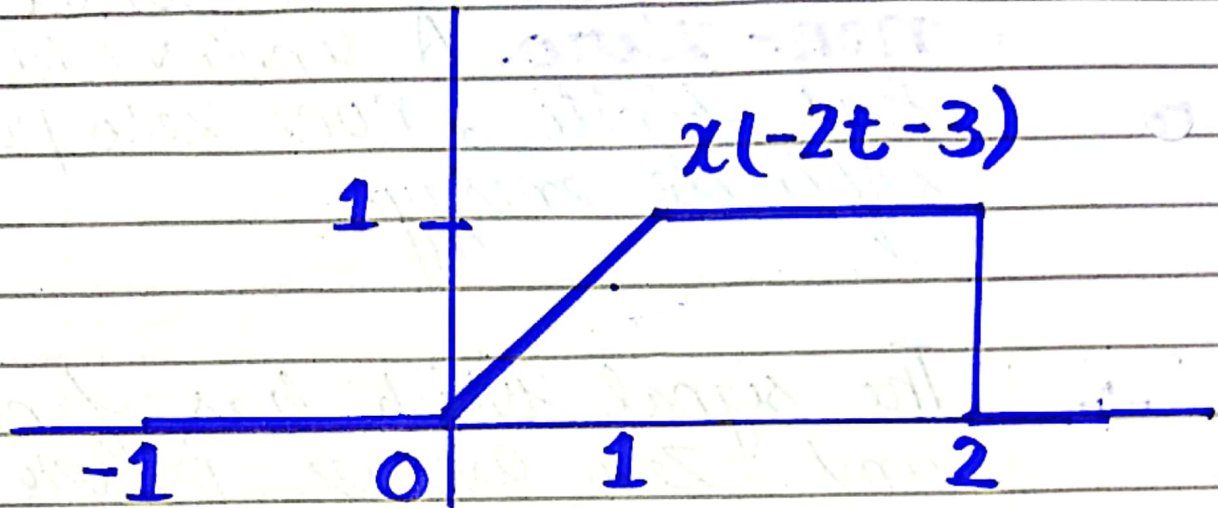
→ The signal which has finite energy and zero average power is called an energy signal.

$$\rightarrow 0 < E < \infty$$

$$\rightarrow \text{and } P=0$$

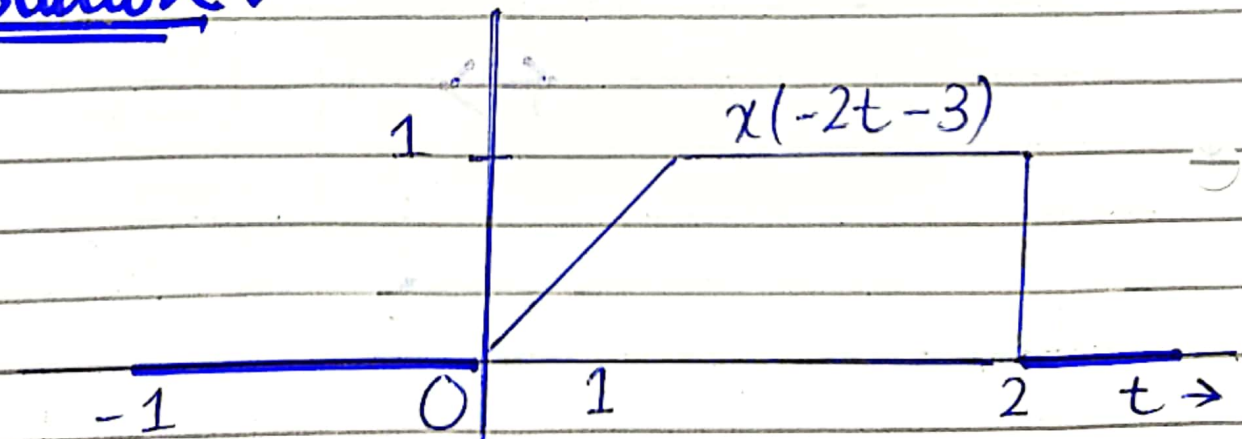


Question # 7



Determine and carefully sketch $x(t)$.

Solution :-



Mathematically:

Step 1:-

→ Time expansion:-

The signal is expanded by factor 2

$$\phi(t) = x\left(\frac{-2t}{2} - 3\right)$$

$$\phi(t) = x(-t - 3)$$

Step 2:-

Time Inversion:-

$$\phi(-t) = x(+t - 3)$$

Step 3:-

$$\phi(t) = x(t - 3)$$

Time advance:-

The signal is advance by factor 3

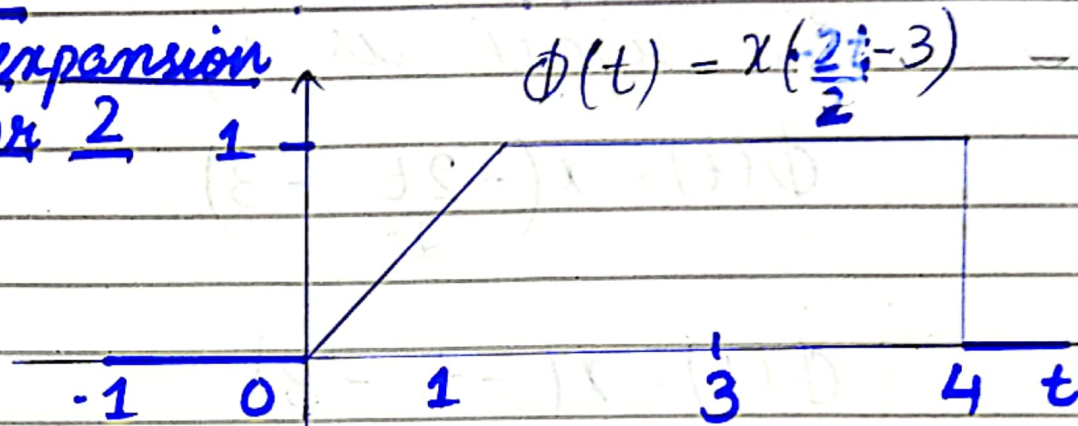
$$\phi(-t + 3) = x(t - 3 + 3)$$

$$\phi(-t + 3) = x(t)$$

Graphically:-

Step 1:-

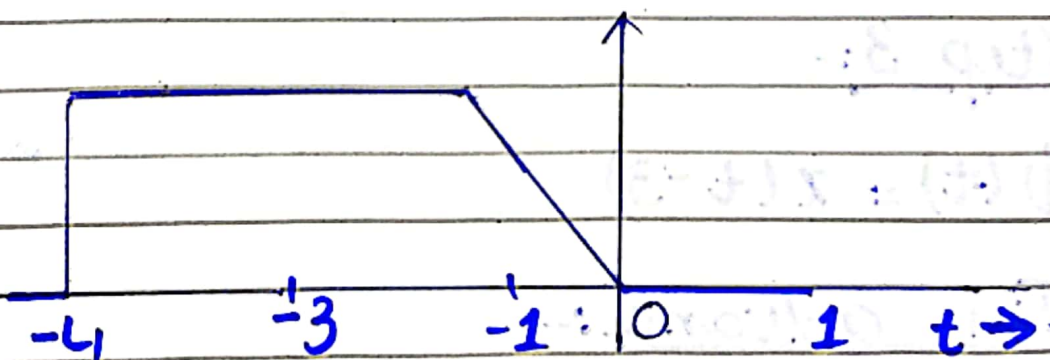
Time expansion
by factor 2



Step 2:-

Time Inversion

$$\phi(-t) = x(+t - 3)$$



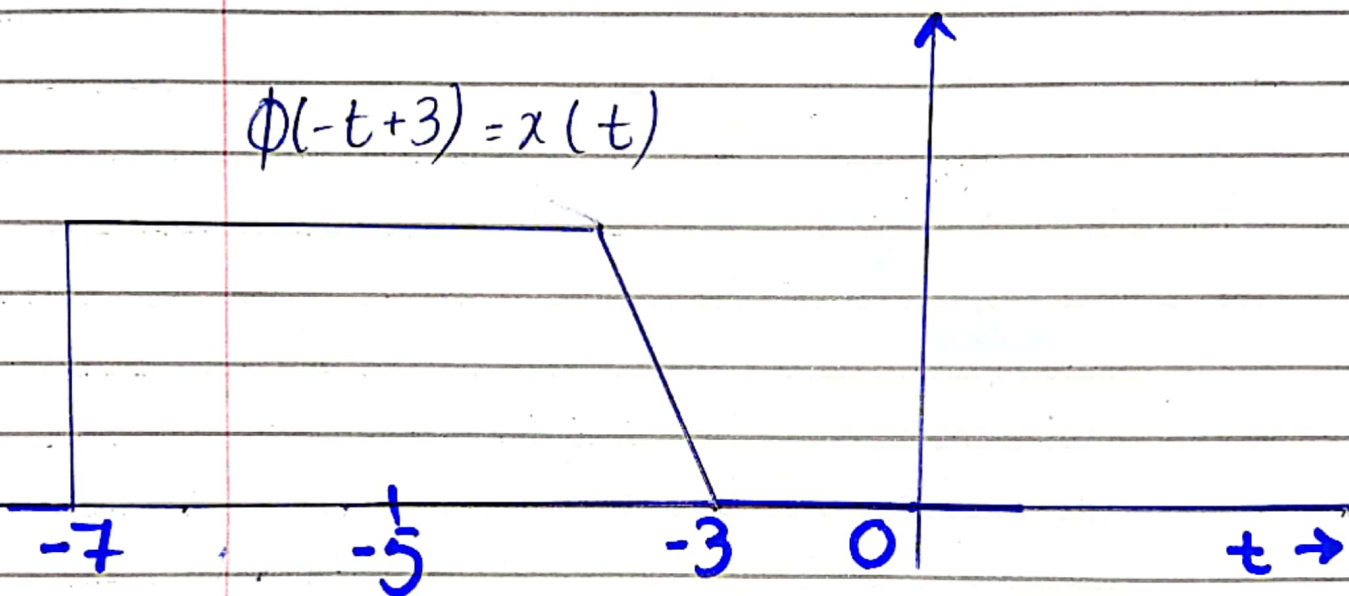
Step 3:-

Time advance:-

$$\phi(-t+3) = x(+t-3+3)$$

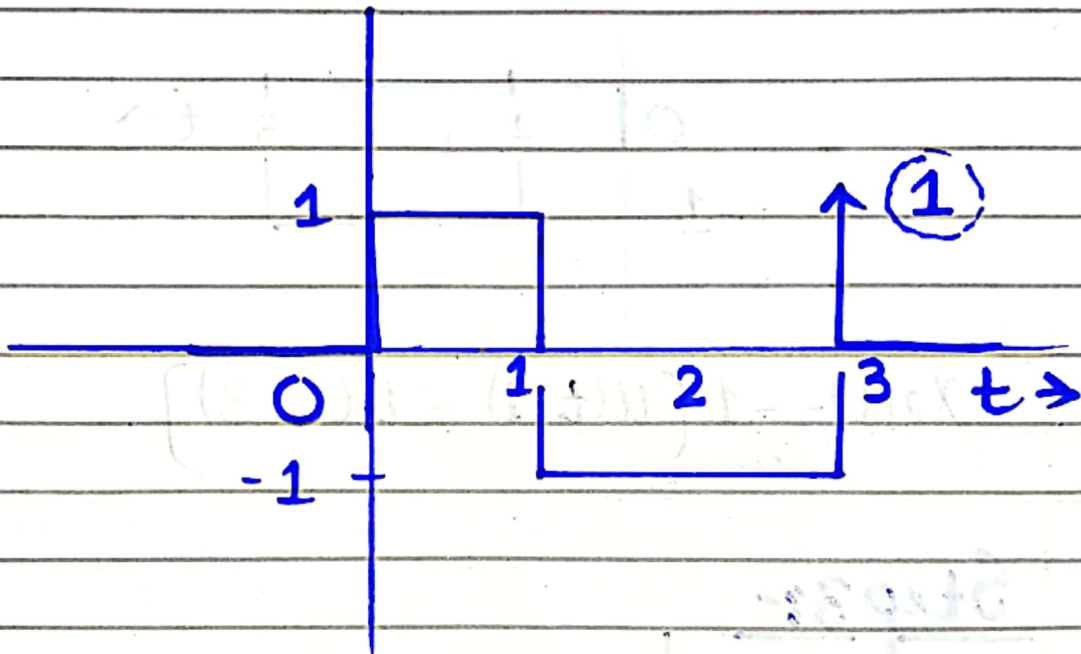
$$\phi(-t+3) = x(t)$$

$$\phi(-t+3) = x(t)$$



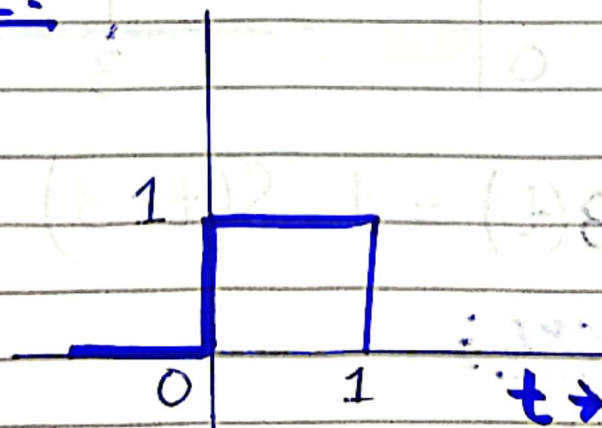
Question #08

find and sketch $\int_{-\infty}^t x(t) dt$



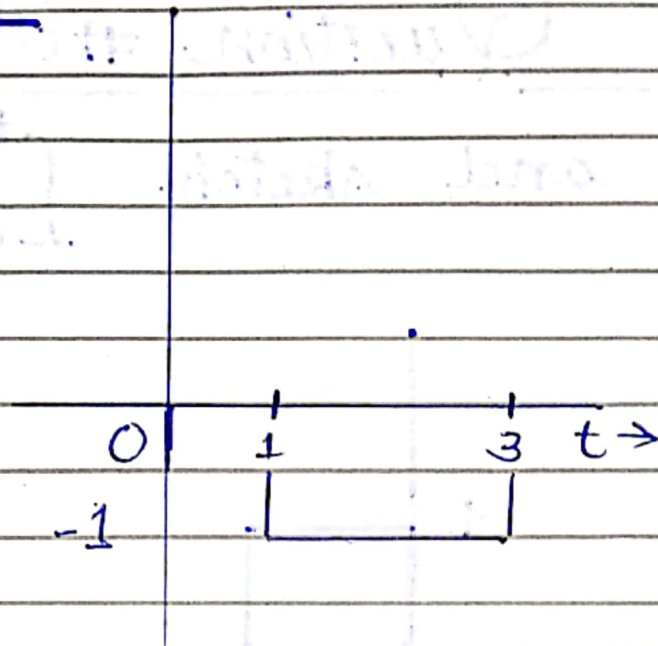
Solution :-

Step 1 :-



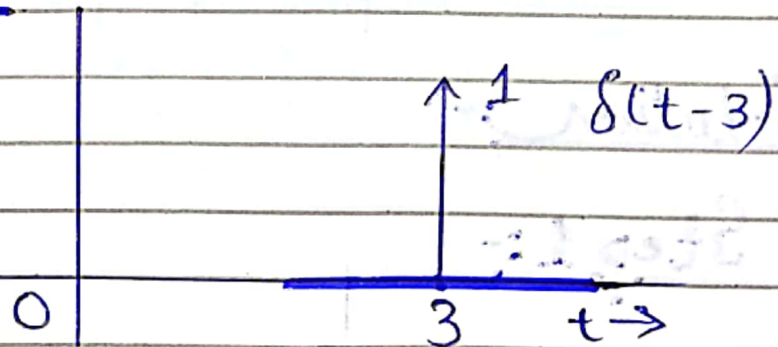
$$x_1(t) = 1[u(t) - u(t-1)]$$

Step 2:-



$$x_2(t) = -1 \cdot [u(t-1) - u(t-3)]$$

Step 3:-



$$x_3(t) = 1 \cdot \delta(t-3)$$

Therefore;

$$x(t) = x_1(t) + x_2(t) + x_3(t)$$

$$x(t) = 1[u(t) - u(t-1)] - 1[u(t-1) - u(t-3)] + [1 \delta(t-3)]$$

$$x(t) = u(t) - u(t-1) - u(t-1) + u(t-3) + \delta(t-3)$$

$$x(t) = u(t) - 2u(t-1) + u(t-3) + \delta(t-3) \quad \rightarrow (1)$$

$\therefore$ Using the concept:-

$$\cdot) \int_{-\infty}^t u(t) dt = x(t)$$

$$\cdot) \int_{-\infty}^t \delta(t) dt = u(t) \quad \nearrow \frac{d u}{dt} = \delta(t)$$

Therefore;

$$\int_{-\infty}^t x(t) = \int_{-\infty}^t [u(t) - 2u(t-1) + u(t-3) + \delta(t-3)]$$

$$\int_{-\infty}^t x(t) = x(t) + x(t-3) - 2x(t-1) + u(t-3)$$

$$\int_{-\infty}^t x(t) = x(t) + x(t-3) - 2x(t-1) + u(t-3) \rightarrow (1)$$

→ for $x(t)$:-

as its co-efficient is "1" (at $t=0$)

$$\Rightarrow m=1$$

→ for $x(t-3)$:-

$$\text{slope} = m = 1 \text{ at } (t=3)$$

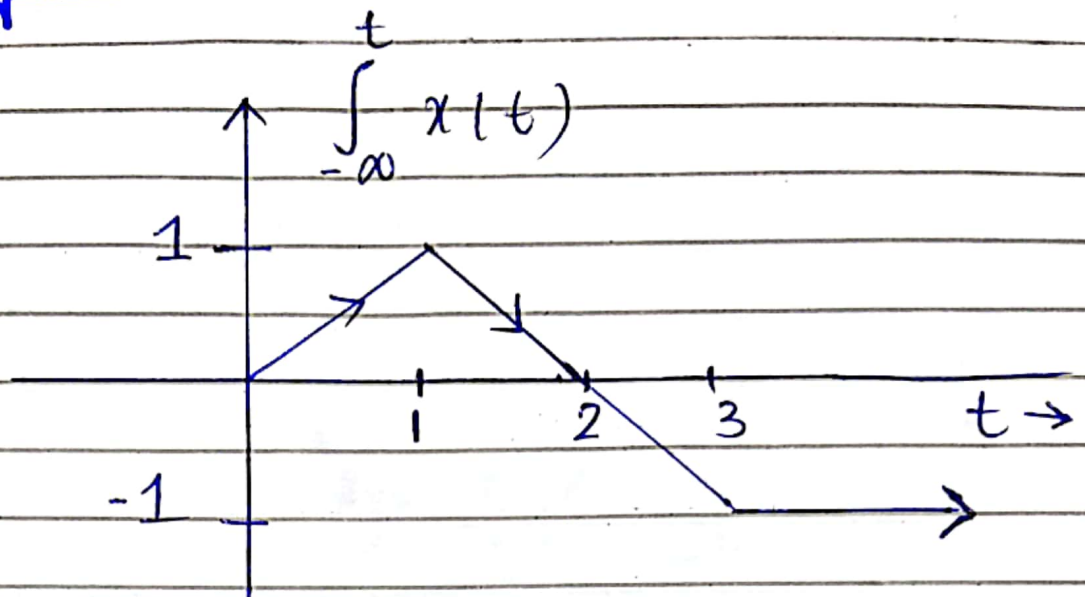
→ for $-2x(t-1)$:-

$$\text{slope} = m = -2 \text{ at } (t=1)$$

→ for $u(t-3)$:-

$$m = 1 \text{ (for } t=3)$$

Therefore;



Question # 09

a) $\frac{dy}{dt} + 2y(t) = x^2(t)$

Determine whether the system described by the equation is linear or non-linear

Solution :-

$$\frac{dy}{dt} + 2y(t) = x^2(t) \rightarrow (1)$$

$\Rightarrow$ Let the system is subjected to the inputs $x_1(t)$ and $x_2(t)$ with response $y_1(t)$ and $y_2(t)$ respectively

For linear system

$$x_1 + x_2 \rightarrow y_1 + y_2 \quad (\text{Additivity})$$

Therefore;

$$\frac{d y_1(t)}{dt} + 2 y_1(t) = x_1^2(t) \rightarrow (a)$$

$$\frac{d y_2(t)}{dt} + 2 y_2(t) = x_2^2(t) \rightarrow (b)$$

Multiplying equation (a) and (b) by the constants K_1 and K_2 respectively

So;

$$K_1 \frac{d y_1(t)}{dt} + K_1 2 y_1(t) = K_1 x_1^2(t) \rightarrow (a_1)$$

$$K_2 \frac{d y_2(t)}{dt} + K_2 2 y_2(t) = K_2 x_2^2(t) \rightarrow (b_1)$$

"Adding" equations (a₁) and (b₁)

$$\therefore K_1 x_1 \rightarrow K_1 y_1$$

$$K_2 x_2 \rightarrow K_2 y_2$$

(homogeneity)

$\Rightarrow$

$$K_1 \frac{dy_1(t)}{dt} + 2y_1(t)K_1 + K_2 \frac{dy_2(t)}{dt} + 2y_2(t)K_2 \\ = K_1 x_1^2(t) + K_2 x_2^2(t)$$

$$\frac{d}{dt} [K_1 y_1(t) + K_2 y_2(t)] + 2[y_1(t)K_1 + y_2(t)K_2] \\ = K_1 x_1^2(t) + K_2 x_2^2(t) \rightarrow (d)$$

As from above eq (d) it is

clear now that system is non-linear because

$$\Rightarrow K_1 x_1^2(t) + K_2 x_2^2(t) \neq K_1 y_1(t) + K_2 y_2(t)$$

for linear system;

$$K_1 x_1^2 + K_2 x_2^2(t) = K_1 y_1^2(t) + K_2 y_2^2(t)$$

b) $\frac{dy}{dt} + 3t y(t) = t^2 x(t)$

Solution :-

$$\frac{dy}{dt} + 3t y(t) = t^2 x(t) \rightarrow (1)$$

Let the system is subjected to the inputs x_1 and x_2 with the responses y_1 and y_2 respectively

$$\therefore x_1 + x_2 \rightarrow y_1 + y_2 \quad \text{(Additivity)}$$

Therefore;

$$\frac{dy_1(t)}{dt} + 3t y_1(t) = t^2 x_1(t) \rightarrow (2)$$

$$\frac{dy_2(t)}{dt} + 3t y_2(t) = t^2 x_2(t) \rightarrow (3)$$

Multiplying equation (2) and (3) by constants K_1 and K_2 respectively

$$\Rightarrow K_1 \frac{dy_1(t)}{dt} + 3tK_1 y_1(t) = t^2 K_1 x_1(t) \rightarrow (4)$$

$$\Rightarrow K_2 \frac{dy_2(t)}{dt} + 3tK_2 y_2(t) = t^2 K_2 x_2(t) \rightarrow (5)$$

Adding equations (4) and (5)

$$K_1 \frac{dy_1(t)}{dt} + 3tK_1 y_1(t) + K_2 \frac{dy_2(t)}{dt} + 3tK_2 y_2(t)$$

$$= t^2 K_1 x_1(t) + t^2 K_2 x_2(t)$$

$$\frac{dy}{dt} [K_1 y_1(t) + K_2 y_2(t)] + 3t [K_1 y_1(t) + K_2 y_2(t)]$$

$$= t^2 [K_1 x_1(t) + K_2 x_2(t)]$$

(linear system)

As,

$$K_1 x_1(t) + K_2 x_2(t) = K_1 y_1(t) + K_2 y_2(t)$$

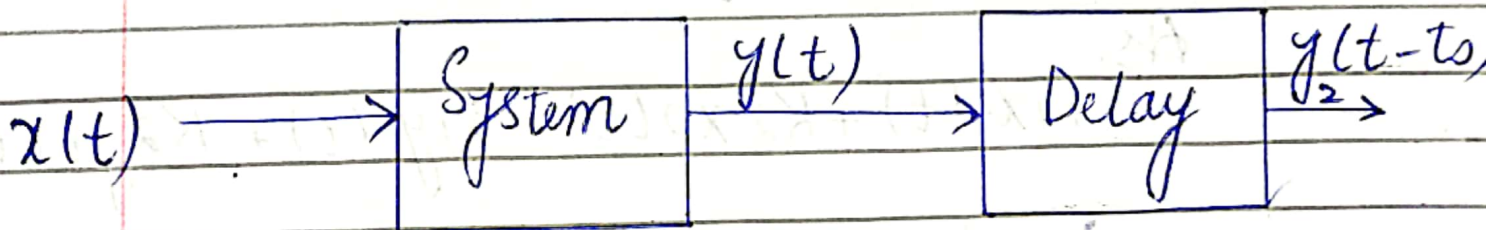
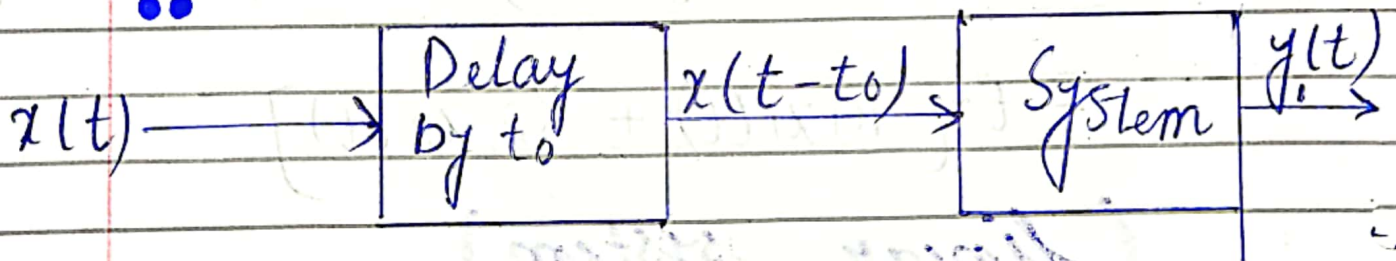
$\Rightarrow$ ignoring constants

Question # 10

→ Determine whether system is time varying or time invariant

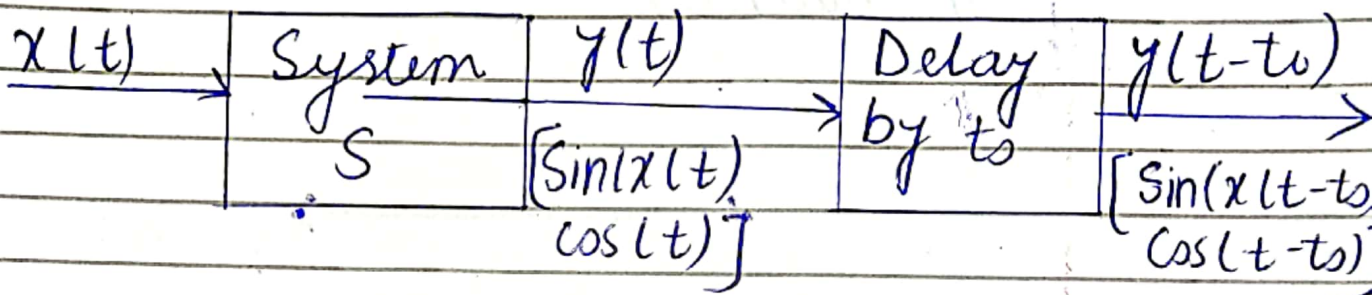
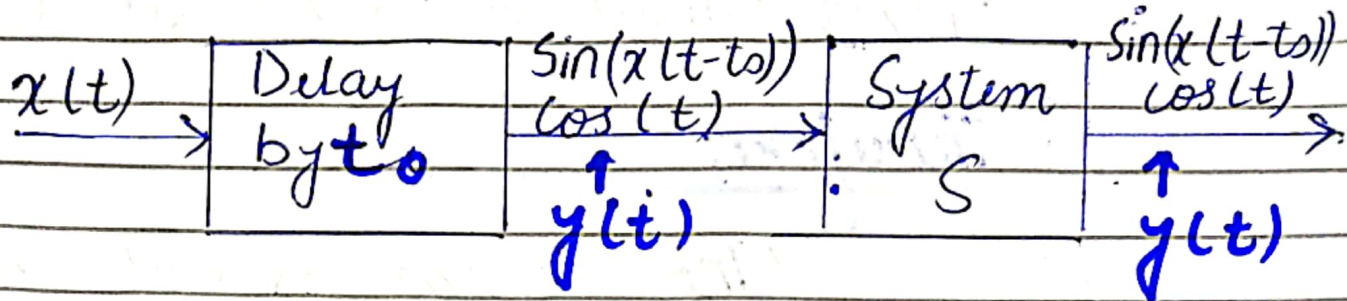
a) $y(t) = \sin(x(t)) \cos t$

⇒ The system is time invariant when changes in input results in identical changes in the output



If $y_1(t) = y_2(t)$ then the system is time invariant

as, $y(t) = \sin(\lambda(t)) \cos(t)$



as from above diagram;

•) $x(t) \xrightarrow{\text{Delay}} \frac{\sin(\lambda(t-t_0))}{\cos(t)} \xrightarrow{S} \frac{\sin(\lambda(t-t_0))}{\cos(t)} \uparrow y_1(t)$

•) $x(t) \xrightarrow{S} \frac{\sin(\lambda(t))}{\cos(t)} \xrightarrow{\text{Delay}} \frac{\sin(\lambda(t-t_0))}{\cos(t-t_0)} \uparrow y_2(t-t_0)$

from above we see ;

$y_1(t) = \frac{\sin(\lambda(t-t_0))}{\cos(t)} \rightarrow (1)$

$y_2(t-t_0) = \frac{\sin(\lambda(t-t_0))}{\cos(t-t_0)} \rightarrow (2)$

$$y_1(t) \neq y_2(t-t_0)$$

$$\sin(x(t-t_0))\cos t \neq \sin(x(t-t_0))\cos(t-t_0)$$

Therefore; :

(The system is not time Invariant)

